

Web Applications --> Java

Core Java

java.net --> ServerSocket, Socket

localhost:8080

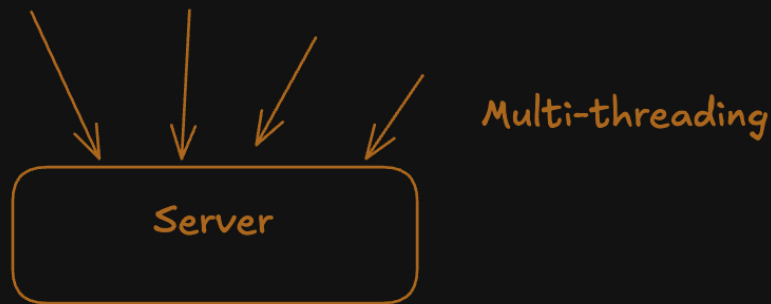
GET /hello
Host: localhost:8080

- HTTP Req

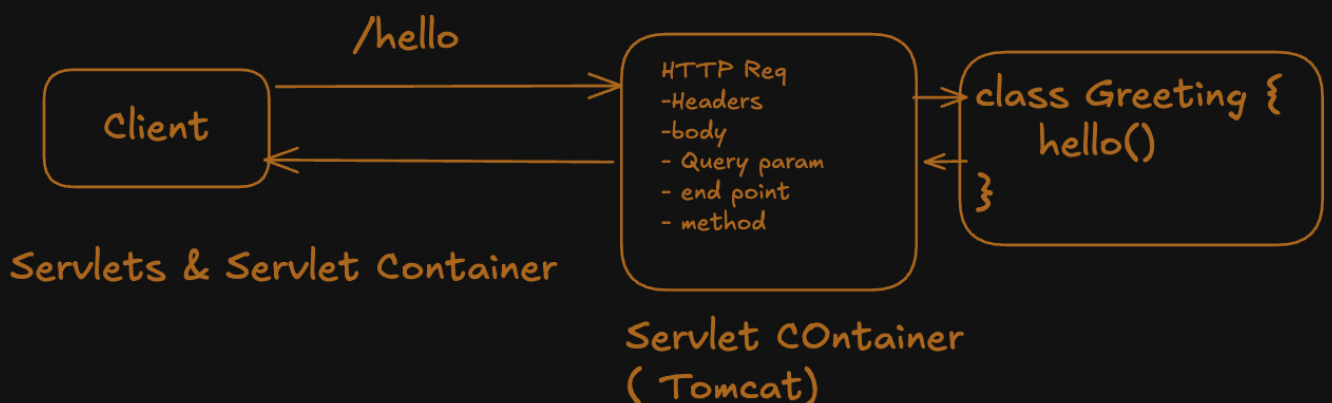
method name --> GET
end point --> /hello
headers -->
body -->

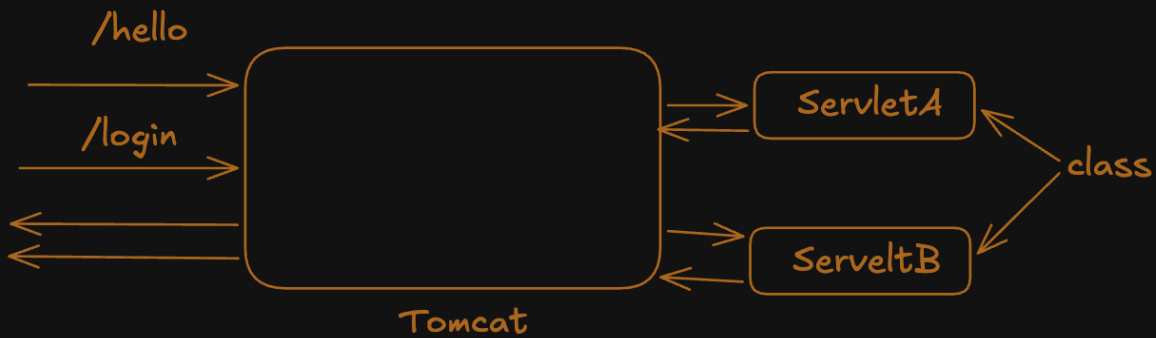
/hello --> hello() {}
/login --> login() {}
/courses --> courses() {}

```
if(endpoint.equals("/hello") && method.equals("GET")) {  
    // call hello() method  
}
```

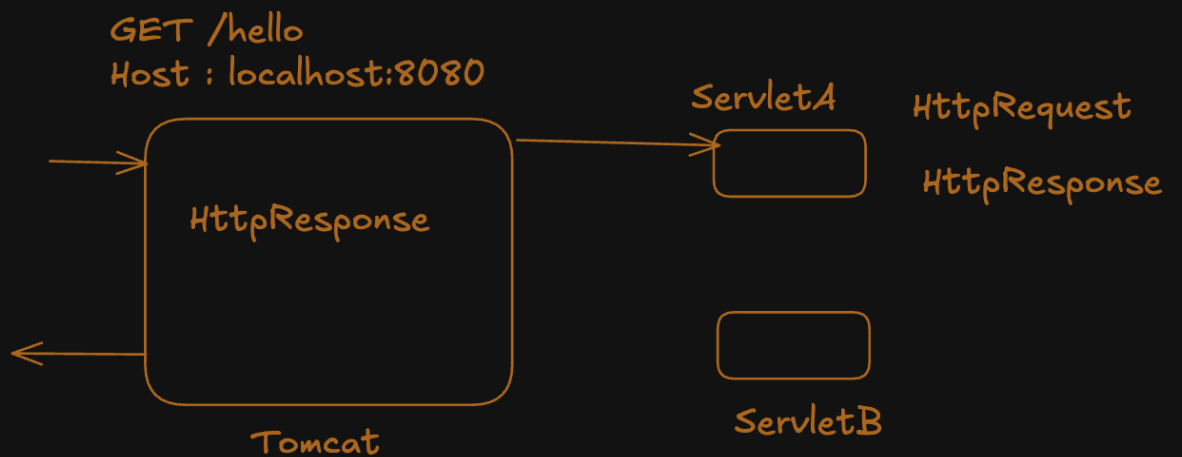
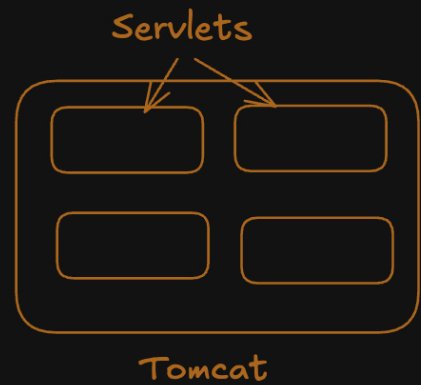
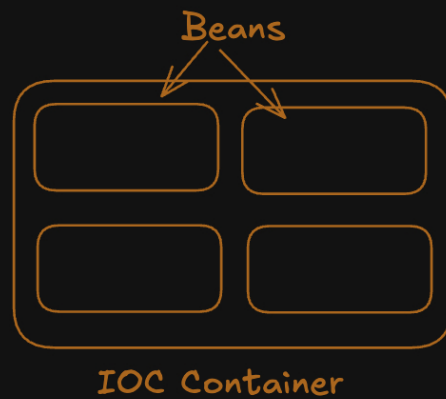


1. Read raw HTTP text
2. Parse the URL
3. Parse headers
4. Parse query parameters
5. Parse request body
6. Identify GET, POST, PUT, DELETE
7. Create response text
8. Add status code
9. Add response headers
10. Send response back to browser
11. Manage multiple users at the same time
12. Manage threads
13. Keep the server running continuously





Java server --> Java web applications
Long running process -> port (8080)



-> External Tomcat

WAR

WAR --> Web Archive

- > compiled class files
- > config
- > Servlet classes
- > HTML/CSS/JS

Create --> POST --> /users

Read --> GET --> /users?id=1

ReadAll -> GET --> /users

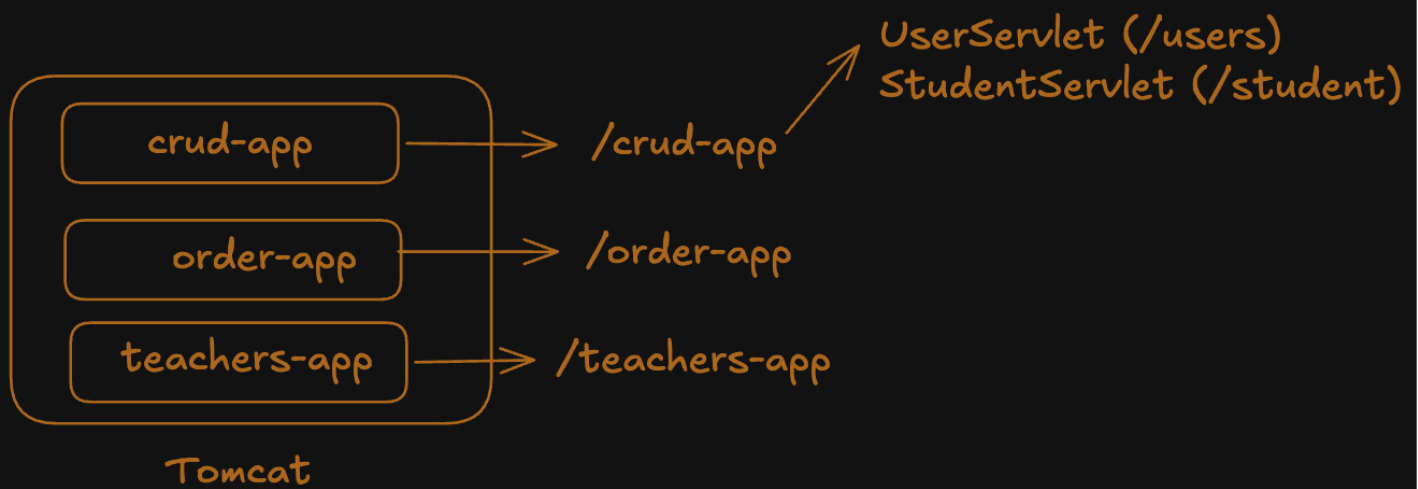
update -> PUT -> /users

Delete -> DELETE -> /users

Tomcat --> POST /users

```
UserService servlet = new UserService();  
servlet.doPost(HttpServletRequest, httpServletRequest);  
  
servlet.doGet();
```

localhost:8080/ crud-app/users



Tomcat --> Servlets

- create
- Destroy

Lifecycle :

init() --> Postconstruct

service() --> doGet(),...

destroy() --> Pre4Destroy



Constructor()

init()

service() --> HttpServlet -->

if(method == GET) doGet()

if(method == POST) doPost()

```
HelloServlet hs = new HelloServlet();  
hs.init();
```

```
hs.service(HttpServletRequest,  
             HttpServletResponse)
```

```
destroy()
```

